

CCNA Course – Practical Project Task

Full Network Design & Implementation in Cisco Packet Tracer

Course:	CCNA (Cisco Certified Network Associate)
Level:	Intermediate
Tool Required:	Cisco Packet Tracer (latest version)
Submission Format:	.pkt file + short PDF/Word report
Estimated Time:	1 – 2 weeks

1. Scenario

You have been hired as a junior network engineer by **NileTech Solutions**, a growing company opening a new branch office. The company has a **Head Office (HQ)** and a **Branch Office** connected over a WAN link. You are required to design, configure, and verify the entire network using Cisco Packet Tracer, applying the concepts covered across the CCNA course.

2. Required Topology

Your network must include the following devices and structure:

- **HQ Site:** 1 Router (HQ-R1), 2 Multilayer/Layer-3-capable Switches, 2 Access Switches, at least 3 VLANs (e.g., Management, Sales, IT), a Server (DHCP + DNS or Web), and at least 6 PCs distributed across VLANs.
- **Branch Site:** 1 Router (BR-R1), 1 Switch, 2 VLANs, and at least 4 PCs.
- **WAN Link:** Connect HQ-R1 and BR-R1 using a serial link (or point-to-point Ethernet if serial modules are unavailable) with a routing protocol.
- **Internet Simulation (optional bonus):** An ISP router providing Internet access via NAT.

3. Configuration Requirements

3.1 VLANs & Switching

- Create at least 3 VLANs at HQ and 2 at Branch with meaningful names.
- Configure trunk links between switches (802.1Q).
- Configure access ports for end devices in correct VLANs.
- Enable port security on at least 2 access ports (max MAC = 2, violation = shutdown).

3.2 Inter-VLAN Routing

- Implement inter-VLAN routing using either Router-on-a-Stick or a Layer-3 switch (SVI).
- Verify that all VLANs can reach each other where permitted by policy.

3.3 IP Addressing

- Design a logical IP addressing scheme using VLSM based on the number of hosts per VLAN/link.
- Fill in the IP Addressing Table (Section 5) before configuring devices.

3.4 Dynamic Routing

- Configure OSPF (single or multi-area) between HQ-R1 and BR-R1 to advertise all networks.
- Verify full routing table convergence using *show ip route* and *show ip ospf neighbor*.

3.5 DHCP & DNS

- Configure a DHCP server (on the Server or router) to automatically assign addresses to at least 2 VLANs.
- Optionally configure DNS name resolution for the internal web/server.

3.6 Access Control Lists (ACLs)

- Create at least one standard or extended ACL to restrict traffic (e.g., block Sales VLAN from reaching Management VLAN, or restrict Telnet/SSH access to the Management VLAN only).
- Document the purpose of every ACL rule you apply.

3.7 Network Address Translation (NAT)

- Configure NAT/PAT on the router facing the simulated ISP (if included) so internal hosts can reach an external/Internet-simulated network.

3.8 Device Security & Management

- Set hostnames, banners (MOTD), and encrypted passwords (enable secret) on all routers/switches.
- Configure SSH (not Telnet) for remote management, with a local username/password and domain name for RSA keys.
- Disable unused ports on all switches.

4. Verification Tasks

Before submitting, verify and be ready to demonstrate the following using appropriate *show* commands and Packet Tracer's Simulation mode:

- End-to-end connectivity between all VLANs and both sites (successful pings) where policy allows it.
- Correct VLAN trunking and port assignments (*show vlan brief*, *show interfaces trunk*).
- OSPF neighbor adjacency and routing table entries (*show ip ospf neighbor*, *show ip route*).
- DHCP is correctly leasing addresses to client PCs (set to DHCP mode).
- ACLs correctly permit/deny traffic as designed (test both allowed and blocked scenarios).
- NAT translation table shows active translations (*show ip nat translations*), if implemented.
- SSH access works from a management PC and Telnet is disabled/refused.

5. IP Addressing Table (to be completed by trainee)

Device	Interface/VLAN	IP Address	Subnet Mask	Default Gateway

6. Deliverables

- The completed Packet Tracer file (.pkt), fully configured and saved.
- A short report (PDF or Word, 2–4 pages) containing: the completed IP addressing table, a network topology diagram/screenshot, and a brief explanation of each design decision (VLAN design, routing protocol choice, ACL logic, NAT setup).
- Screenshots of at least 3 verification commands proving the network works as intended.

7. Grading Rubric

Criteria	Points
Topology design & device connections	10
VLAN configuration & trunking	15
IP addressing scheme (VLSM) & accuracy	15

Criteria	Points
Inter-VLAN routing	10
OSPF configuration & convergence	15
DHCP configuration	10
ACL design & correctness	10
NAT configuration	5
Device security (SSH, passwords, banners, port security)	5
Report quality & documentation	5
Total	100

8. Submission Notes

Rename your files as **StudentName_CCNA_FinalProject.pkt** and **StudentName_CCNA_FinalProject_Report.pdf** before submission. Late submissions and plagiarized/copied topologies will be penalized per course policy.

Good luck – design it as if this network will actually go live.